

Ningna Wang

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My research interests broadly lie in **computer graphics**, **geometry processing**, and **3D shape analysis**. My current research direction focuses on **3D meshing**, **3D reconstruction** and **3D shape analysis**. Additionally, I am also interested in exploring **aerial path planning** for 3D urban scene reconstruction and **generative 3D shape synthesis**.

Research Positions

Postdoctoral Research Scientist, Columbia University New York, NY, USA
Advisor: Dr. Silvia Sellán [↗](#) 2025 - now

- Developing a series of robust methods to **extract surface mesh** from implicit fields.

PhD Student, The University of Texas at Dallas Dallas, Texas, USA
Advisor: Dr. Xiaohu Guo [↗](#) 2019 - 2025

- Developed a complete framework for computing **high-quality** medial meshes of 3D models with **structure-awareness**.
- Developed a complete framework for computing the medial axis of 3D CAD meshes with **sharp-features preservation**.
- Developed a novel **topology-preserving** 3D medial axis computation framework based on volumetric restricted power diagram (RPD).
- Researched on new **learning-based methods** for mesh reconstruction via 3D skeleton prediction from **single view images** or **point clouds**.
- Developed a new method for estimating **globally consistent normal orientations** for a raw point cloud.
- Studied a smooth **mesh simplification** functional that simultaneously consolidates weak features in a high-quality mesh.

Research Intern, Shenzhen University Shenzhen, Guangdong, China
Advisor: Dr. Hui Huang [↗](#) 10/2023 - 12/2023

- Developed an **aerial path planning** method for high-quality digital 3D scene updation.

Degrees

Ph.D. in Computer Science	University of Texas at Dallas Dallas, Texas, USA	2025
M.S. in Computer Science	Carnegie Mellon University Pittsburgh, PA, USA	2016
B.S. in Computational Mathematics	Jilin University Changchun, Jilin, China	2014

Awards

2025	EECS Rising Stars ↗	Massachusetts Institute of Technology (MIT)	USA
2025	Rising Stars in Computer Graphics ↗	ACM WiGRAPH	USA
2023	SIGGRAPH Technical Best Paper Award ↗	ACM SIGGRAPH	USA
2025	CVM 2025 Best Paper Award ↗	Computational Visual Media Conference	HK, China
2024	Betty and Gifford Johnson Travel Awards	The University of Texas at Dallas	USA
2024	First Place CAST STAR Award (1/16 teams)	CAST-STEM Bridge Summer Camp	USA
2013	Honorable Mention	Mathematical Contest in Modeling (MCM)	USA
2010	First/Second Prize Scholarship & Outstanding Student	Jilin University (2010-2014)	China

Publications

[JOURNAL & CONFERENCE ARTICLES] * denotes equal contributions

- [1] MATStruct: High-quality Medial Mesh Computation via Structure-aware Variational Optimization
[Ningna Wang](#), Rui Xu, Yibo Yin, Zichun Zhong, Taku Komura, Wenping Wang, Xiaohu Guo
ACM SIGGRAPH Asia. Hong Kong, China, 2025: pp. 1–12.
- [2] MATTopo: Topology-preserving Medial Axis Transform with Restricted Power Diagram [Journal]
[Ningna Wang](#), Hui Huang, Shibo Song, Bin Wang, Wenping Wang, Xiaohu Guo
ACM Transactions on Graphics, 2024, 43(6): pp. 1–16.
- [3] Computing Medial Axis Transform with Feature Preservation via Restricted Power Diagram [Journal]
[Ningna Wang](#), Bin Wang, Wenping Wang, Xiaohu Guo

ACM Transactions on Graphics, 2022, 41(6): pp. 1–18.

- [4] Aerial Path Online Planning for Urban Scene Updation
Mingfeng Tang*, Ningna Wang*, Ziyuan Xie, Jianwei Hu, Ke Xie, Xiaohu Guo, Hui Huang
ACM SIGGRAPH North America. Vancouver, Canada, 2025: pp. 1–11.
- [5] Dual Contouring of Signed Distance Data
Xiana Carrera, Ningna Wang, Christopher Batty, Oded Stein, Silvia Sellán
ACM SIGGRAPH North America. Los Angeles, USA, 2026.
- [6] Globally Consistent Normal Orientation for Point Clouds by Regularizing the Winding-Number Field **[Best Paper Award]**
Rui Xu, Zhiyang Dou, Ningna Wang, Shiqing Xin, Shuangmin Chen, Mingyan Jiang, Xiaohu Guo, Wenping Wang, Changhe Tu
ACM Transactions on Graphics, 2023, 42(4): pp. 1–15.
- [7] CWF: Consolidating Weak Features in High-quality Mesh Simplification **[Journal]**
Rui Xu*, Longdu Liu*, Ningna Wang, Shuangmin Chen, Shiqing Xin, Xiaohu Guo, Zichun Zhong, Taku Komura, Wenping Wang, Changhe Tu
ACM Transactions on Graphics, 2024, 43(4): pp. 1–14.
- [8] MVFusion-GS: Motion-Variance Guided Temporal Attention for High-Quality Dynamic Gaussian Splatting
Jianwei Hu, Tingxuan Huang, Hengyu Zhou, Ningna Wang, Xiaohu Guo Jinshan Lai, Bin Wang
European Conference on Computer Vision (ECCV), 2026.
- [9] NASM: Neural Anisotropic Surface Meshing
Hongbo Li, Haikuan Zhu, Sikai Zhong, Ningna Wang, Cheng Lin, Xiaohu Guo, Shiqing Xin, Wenping Wang, Jing Hua, Zichun Zhong
ACM SIGGRAPH Asia. Tokyo, Japan, 2024: pp. 1–12.
- [10] CADDreamer: CAD Object Generation from Single-view Images
Yuan Li, Cheng Lin, Yuan Liu, Xiaoxiao Long, Chenxu Zhang, Ningna Wang, Xin Li, Wenping Wang, Xiaohu Guo
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). Nashville, USA, 2025.
- [11] TexHOI: Reconstructing Textures of 3D Unknown Objects in Monocular Hand-Object Interaction Scenes **[Journal]**
Alakh Aggarwal, Ningna Wang, Xiaohu Guo
IEEE Transactions on Visualization & Computer Graphics (TVCG), 2025, 31(10): pp. 8347–8358.
- [12] S3DS: Self-supervised Learning of 3D Skeletons from Single View Images
Jianwei Hu, Ningna Wang, Baorong Yang, Gang Chen, Xiaohu Guo, Bin Wang
Proceedings of the 31st ACM International Conference on Multimedia (ACM MM). Ottawa, Canada, 2023: pp. 6948–6958.
- [13] RBF-MAT: Computing medial axis transform from point clouds by optimizing radial basis functions **[Journal]**
Mengyuan Ge, Junfeng Yao, Baorong Yang, Ningna Wang, Zhonggui Chen, Xiaohu Guo
Computer Aided Geometric Design (CAGD), 2025, 118(C): p. 102421.
- [14] Point2MM: Learning medial mesh from point clouds **[Journal]**
Mengyuan Ge, Junfeng Yao, Baorong Yang, Ningna Wang, Zhonggui Chen, Xiaohu Guo
Computers & Graphics (C&G), 2023, 115(C): pp. 511–521.
- [15] IMMAT: Mesh Reconstruction from Single View Images by Medial Axis Transform Prediction **[Journal]**
Jianwei Hu, Gang Chen, Baorong Yang, Ningna Wang, Xiaohu Guo, Bin Wang
Computer-Aided Design (CAD), 2022, 150: p. 103304.

[IN SUBMISSIONS]

- [1] VoroUDF: Meshing Unsigned Distance Fields with Voronoi Optimization **[Journal]**
Ningna Wang, Zilong Wang, Xiana Carrera, Xiaohu Guo, Silvia Sellán
submitted to ACM Transactions on Graphics, 2026.
- [2] Superpower Contouring of Unsigned Distance Data **[Journal]**
Ningna Wang, Xiana Carrera, Christopher Batty, Oded Stein, Silvia Sellán
submitted to ACM Transactions on Graphics, 2026.
- [3] CSG-Agent: CSG Recovery and Simplification for B-rep CAD Models **[Journal]**
Ang Zhao, Yuan Li, Ningna Wang, Bingbing Hu, Junfeng elder Yao, Hao Pan, Congyi Zhang, Xiaohu Guo
submitted to ACM Transactions on Graphics, 2026.
- [4] DC-SQEM: Medial Mesh Extraction by Dual Contouring with Spherical Quadric Error Metric **[Journal]**
Bingchuan Li, Ruilong Liu, Tianhao Hu, Congyi Zhang, Ningna Wang, Junfeng Yao, Xiaohu Guo
submitted to ACM Transactions on Graphics, 2026.

Invited Talks

Contouring and Operations on Implicit Geometry

nTop [↗](#) | [New York, NY, USA](#) 06/2026

Inside Shapes: Understanding Shapes through Medial Geometry

University of Toronto | [Toronto, Canada](#) 04/2026

New England Symposium on Graphic [↗](#) | [MIT, Cambridge, MA, USA](#) 04/2026

GAMES Webinar | [Online](#) 04/2026

Computing 3D Medial Axis Transform via Restricted Power Diagram (RPD)

Visual Computing Research Center, Shenzhen University | [Shenzhen, China](#) 12/2025

MATStruct: High-Quality Medial Mesh Computation via Structure-aware Variational Optimization

ACM SIGGRAPH Asia 2025 | [HongKong, China](#) 12/2025

Center for Digital Media Computing, Xiamen University | [Online](#) 12/2025

Aerial Path Online Planning for Urban Scene Updation

ACM SIGGRAPH 2025 | [Vancouver, Canada](#) 08/2025

MATTopo: Topology-preserving Medial Axis Transform with Restricted Power Diagram

ACM SIGGRAPH ASIA 2024 | [Tokyo, Japan](#) 12/2024

Visual Computing Research Center, Shenzhen University | [Shenzhen, China](#) 11/2023

Center for Digital Media Computing, Xiamen University | [Online](#) 10/2024

Computing Medial Axis Transform with Feature Preservation via Restricted Power Diagram

ACM SIGGRAPH ASIA 2022 | [Daegu, South Korea](#) 12/2022

Center for Digital Media Computing, Xiamen University | [Online](#) 01/2024, 11/2022

Other Work Experience

Booking.com B.V.

[Amsterdam, Netherlands](#)

Senior Software Engineer 2018 - 2019

Software Engineer 2017 - 2018

- [System Design and Development] Responsible for the design, development, and continued operation of the **hotel availability search system**, which handles thousands of incoming hotel search requests per second.
- [Production Infrastructure Optimization] Significantly enhanced system stability and scalability by distributing hotel availabilities using **jump consistent hashing**, a fast consistent hash algorithm with no storage and minimal memory requirements.
- [Cross-Functional Collaboration] Collaborated seamlessly with product-side engineers and partner-side engineers to ensure the successful development and delivery of the search system.

The Priceline Group Inc.

[Amsterdam, Netherlands](#) | [Seattle, WA, USA](#)

Graduate Software Engineer 2016 - 2017

- [System Design] Developed a **hotel inventory management system** with a wealth of features, including property listing, yield management, and revenue analytics.
- [Feature Optimization] Implemented and experimented new features for the **Genius loyalty program** for various discounts and travel rewards.

Teaching

Mentor MIT Summer Geometry Initiative [↗](#) 2026 Summer

- Mentor of a 5-undergrad-students team on a 1-week learning-based 3D cross-sections reconstruction research project.

Teaching Assistant UTD CS4361/CS6366 Computer Graphics 2025 Spring, 2024 Fall, 2021 Spring

Teaching Assistant UTD Clark Summer Research Program 2024 Summer

Teaching Assistant CAST-STEM Bridge Summer Camp [↗](#) 2024 Summer

- Supervisor of a 9-high-school-students team on a 6-weeks 3D talking face project.
- Instructor of 2-weeks full-day deep learning tutorials including lectures, coding demos and exercises.

Teaching Assistant UTD CS6323 Computer Animation and Gaming 2022 Fall

Teaching Assistant UTD CS4347 Database Systems 2022 Fall, 2021 Spring

Teaching Assistant UTD CS6334 Virtual Reality 2020 Spring

Teaching Assistant UTD CS4332 Introduction to Programming Video Games 2019 Fall

Service

Conference	Committee Member	ACM SIGGRAPH North America	2027 (expected)
Conference	Posters Jury ↗	ACM SIGGRAPH North America	2026
Conference	Reviewer	ACM SIGGRAPH North America	2026, 2025, 2024
Conference	Reviewer	ACM SIGGRAPH Asia	2026, 2025, 2024
Journal	Reviewer	ACM Transactions on Graphics (TOG)	2026, 2025, 2024
Journal	Reviewer	IEEE Transactions on Visualization and Computer Graphics (TVCG)	2026, 2025, 2022
Journal	Reviewer	Computer Aided Geometric Design (CAGD)	2024
Journal	Reviewer	Graphical Models (GMOD)	2025, 2024
Conference	Program Committee	Symposium on Solid and Physical Modeling (SPM)	2026
Conference	Program Committee	Computer Graphics International (CGI)	2025
Conference	Program Committee	International Conference on Computational Visual Media (CVM)	2024
Conference	Program Committee	Pacific Graphics	2026
Conference	Reviewer	Pacific Graphics	2023
Conference	Reviewer	International Conference on 3D Vision (3DV)	2026
Conference	Reviewer	International Conference on Geometric Modeling and Processing (GMP)	2024

Student Mentorship

Chronologically by start date.

Xiana Carrera ↗	PhD candidate at Columbia University (USA)	2025-now
Bea Stotz	Undergrad at Columbia University (USA)	2025-now
Bingchuan Li	PhD candidate at Xiamen University (China)	2025-now
Ruilong Liu	MS at Xiamen University 2025, PhD candidate at Xiamen University (China)	2025-now
Tianhao Hu	MS student at Xiamen University (China)	2025-now
Zilong Wang ↗	PhD candidate at The University of Texas at Dallas (USA)	2025-now
Mingfeng Tang	MS student at Shenzhen University (China)	2024-now
Ziyuan Xie	MS student at Shenzhen University (China)	2024-2025
Yibo Yin ↗	BS at Wuhan University (China) 2025; MS student at EPFL (Switzerland)	2024-2025, 2026-now
Yuan Li	PhD candidate at The University of Texas at Dallas (USA)	2024-2025
Mengyuan Ge	PhD candidate at Xiamen University (China)	2022-2024
Shibo Song ↗	MS at Xiamen University (China) 2023; After graduation: miHoYo	2021-2024

Software

Additionally, open-source code is available for publications above at <https://ningnawang.github.io/>.

LibMAT

github.com/ningnawang/libmat [↗](#)

A C++ library of data structure and algorithms for processing the 3D medial axis transform (MAT), with a special emphasis on medial meshes.

Blender-mat-addon

github.com/songshibo/blender-mat-addon [↗](#)

A Blender plugin that enables import and export of medial meshes within Blender. It supports loading medial meshes from MA format files and can interpolate all medial primitives, including spheres, cones, and slabs.

Q-MAT open-source

github.com/ningnawang/QMAT [↗](#)

An open-source C++ code repository for SIGGRAPH 2015 paper *Q-MAT: Computing Medial Axis Transform by Quadratic Error Minimization*. The code is compatible with both macOS and Windows platforms.

References available upon request.